

Article

Trends in Non-Profit Cybersecurity: Analyzing Three Years of Incident Data from the NPCIR

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Abstract

This study analyzes cyberattack trends targeting non-profit organizations using longitudinal data collected over a three-year period within the Non-Profit Cybersecurity Incident Repository (NPCIR). Developed through a National Security Agency Center of Academic Excellence in Cyber Defense (NSA CAE-CD) designated center, the NPCIR applies an open-source intelligence (OSINT) methodology to systematically document cybersecurity incidents affecting the global non-profit sector. This study examines attack types, threat actor characteristics, sectoral distribution, and cybersecurity impacts using the Confidentiality–Integrity–Availability (CIA) triad framework. The results indicate that availability-related incidents, particularly ransomware and distributed denial-of-service (DDoS) attacks, constitute the most prevalent threats, while confidentiality breaches remain highly significant due to frequent data exposure incidents. Statistical analyses further demonstrate significant differences between non-profit organizations aligned with DHS CISA critical infrastructure sectors and those operating outside these sectors, especially regarding the prevalence of availability-focused attacks. In addition to its empirical contribution, the NPCIR initiative supports experiential learning opportunities for undergraduate and graduate students in cybersecurity and information technology. The resulting dataset provides actionable cyber threat intelligence for researchers, practitioners, and non-profit leaders seeking to strengthen organizational cybersecurity resilience and awareness.

Keywords: non-profit organizations; cybersecurity; attack types; cyberattack trends

1. Introduction

Non-profit organizations remain a staple and enduring presence in most parts of the globe. In many cases, non-profits fill an unmet need or void that the local, state, or federal government agencies are not fulfilling to help communities. An effort was started several years back to catalog cybersecurity and cybercrime attacks against this sector, which included collected data related to attack types, threat actors, and critical infrastructure-aligned sectors [1]. In this successive effort, further analysis and data are provided on this longitudinal, multi-variable resulting dataset of real-world cyber incidents against the non-profit arena. Non-profit organizations continue to be challenged by inadequate cybersecurity practices and policies, which are attributed to resource limitations as well as lacking expertise and awareness [2–4]. This research aims to further contribute to greater threat intelligence awareness for the non-profit sector. The effort to collect and catalog open-source intelligence information relating to non-profits is ongoing and is expected



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to continue in the future as part of this project. This paper provides a snapshot of where the data have provided valuable insights. Included in this paper are the theoretical model followed, a literature review, results in the current instantiation of the longitudinally collected data, a discussion, and a limitations section that rounds out the paper.

Theoretical Model and Framework

Several theoretical models were considered in this work. These included Routine Activity Theory (RAT), the Socio-Technical Systems Theory (STST), Organizational Resilience Theory, and Opportunity Theory. Ultimately, Routine Activity Theory, coupled or extended with Socio-Technical Systems Theory, was applied in this research. Routine Activity Theory, targeted at the non-profit cybersecurity and threat landscape, augmented by Socio-Technical Systems Theory, provides an explanation of how organizational structure and technical security design can together help shape cyberattack victim patterns.

The core idea of Routine Activity Theory is that a crime occurs when a motivated offender targets a suitable endpoint, and there is an absence of a capable guardian [5]. Within the realm of this research, the motivated offenders included threat actors such as cybercriminals, advanced persistent threat (APT) groups, hacktivists, or script kiddies. The targets included any non-profit organization, regardless of geography, that focuses on a variety of mission areas or sectors. The lack of a guardian can be evidenced by recurring attacks or breaches, systems that remain vulnerable through unpatched hardware or software, and an example of success in ransomware attacks.

With regard to Socio-Technical Systems Theory (STS), technical systems in the form of computer and networking hardware and software interact with the social or human side of things, including processes, procedures, oversight, management, and governance. Bostrom and Heinen [6,7] initiated and developed STS, which provides the framing with information systems, cybersecurity, and other solution integrations, which are the elements of two correlated systems: social and technical constructs. Social constructs integrate well into this research, where emphasis is given to the human investigation of open-source intelligence. The study findings and activities show that many cyber breaches or attacks are not only technical but are rarely purely technical, since attacks can be discovered when systems are not properly configured, end users fall prey to phishing attacks, as well as social engineering attacks, which align with the social side of operations.

2. Literature Review

2.1. Cybersecurity in the Non-Profit Sector

Non-profit organizations constitute a highly diverse and globally distributed sector, operating across a wide range of domains including healthcare, education, humanitarian aid, governance, and social services. Their structural diversity and mission-driven orientation contribute to their societal importance but also introduce variability in their technological maturity and cybersecurity preparedness. As non-profit organizations continue to expand their digital footprint, particularly through the adoption of cloud services, online platforms, and emerging technologies such as artificial intelligence and generative AI, their exposure to cyber threats has increased significantly. The growing reliance on digital tools has introduced new operational efficiencies but also amplified vulnerabilities related to data protection, system security, and incident response capabilities [8]. At the same time, non-profits face persistent challenges related to insufficient disclosure of cyber incidents and limited transparency in reporting security breaches. Unlike heavily regulated sectors, such as finance or energy, the non-profit sector is generally not subject to mandatory reporting requirements, resulting in fragmented and incomplete visibility into the true scale of cyber threats. The scale of the sector further compounds these challenges. Estimates

indicate that more than one million non-profit organizations operate in the United States alone, employing a substantial workforce and managing large volumes of sensitive data [9]. These data often includes donor information, financial records, and, in some cases, highly sensitive healthcare data, making non-profits attractive targets for malicious actors seeking financial gain, data exploitation, or reputational disruption.

The recent literature characterizes non-profit organizations as “cyber-poor but target-rich,” highlighting a structural imbalance between the value of the assets they manage and the level of cybersecurity investment they can sustain [10]. Empirical studies suggest that up to 60% of non-profits have experienced a cyberattack or data breach in recent years, underscoring the widespread nature of the threat [9,11]. Despite this high prevalence, many organizations lack the financial resources, technical expertise, and standardized procedures necessary to effectively prevent, detect, and respond to cyber incidents.

2.2. *Types of Cyberattacks and Threat Actors*

The broader cybersecurity literature identifies several dominant categories of cyberattacks, including data breaches, ransomware, distributed denial of service (DDoS), phishing, and advanced persistent threats (APTs) [12]. These attack types differ in their technical complexity, objectives, and impact on organizational assets. Ransomware has emerged as a particularly disruptive form of attack, especially for organizations with limited recovery capabilities. By encrypting critical systems and data, ransomware primarily affects the availability of services, often resulting in operational paralysis. In contrast, data breaches target confidentiality, enabling unauthorized access to sensitive information that can be monetized or exploited. Threat actors targeting non-profits are diverse and include opportunistic cybercriminals, organized crime groups, hacktivists, and state-sponsored entities. Hacktivism is especially relevant in the non-profit context, as organizations engaged in politically or socially sensitive activities may be targeted due to ideological opposition. This diversity of threat actors increases the complexity of cybersecurity risk management in the sector.

2.3. *The CIA Triad as a Framework for Cybersecurity Analysis*

The Confidentiality, Integrity, and Availability triad remains a foundational framework for analyzing cybersecurity risks and impacts. It enables the structured classification of attack consequences and supports comparative analysis across sectors and organizational types. In non-profit environments, the relative importance of each CIA component may vary depending on the organization’s mission and operational context [13]. For instance, healthcare-related non-profits may prioritize confidentiality due to regulatory and ethical considerations, while humanitarian organizations may emphasize availability to ensure uninterrupted service delivery. Existing research suggests a growing prevalence of attacks targeting availability, reflecting a broader shift toward disruption-oriented cybercrime. However, confidentiality breaches remain highly significant due to the financial and strategic value of stolen data. Integrity-related attacks, although less frequent, can undermine trust and damage organizational reputation, particularly in cases involving fraud or content manipulation.

2.4. *Open-Source Intelligence (OSINT) in Cybersecurity Research*

Open-source intelligence has become an increasingly important methodology for collecting cybersecurity incident data, particularly in sectors where formal reporting mechanisms are limited [14]. OSINT allows researchers to aggregate information from publicly available sources such as news reports, security advisories, and organizational disclosures. This approach is particularly relevant for the non-profit sector, where under-reporting of incidents is common. However, OSINT-based research is subject to several limitations,

including potential bias toward high-profile incidents, inconsistencies in data quality, and challenges with verification. Smaller organizations and less-visible incidents may be under-represented, leading to incomplete datasets. Despite these challenges, the development of structured repositories, such as the Non-Profit Cyber Incident Repository (NPCIR), represents a significant advancement in enabling longitudinal and data-driven analysis of cyber threats affecting non-profit organizations [15].

The literature highlights the increasing vulnerability of non-profit organizations to cyber threats, driven by resource constraints, expanding digital infrastructures, and diverse threat actor motivations. While frameworks such as the CIA triad and OSINT provide valuable analytical tools, there remains a significant need for longitudinal, theory-driven, and data-intensive research. This study contributes to addressing these gaps by leveraging a multi-year dataset and integrating theoretical perspectives to analyze cyberattack trends in the global non-profit sector.

3. Materials and Methods

Collection of the analyzed data has been an ongoing task assigned to the Kean Center for Cybersecurity staff, including graduate assistants, student interns, and faculty. The cybersecurity incidents are found through open sources, and, after the verification and validation of the data, the information is added using a set of fields and a taxonomy. The fields and categories of information collected are outlined in a previous paper that focuses on the feasibility of such an effort [1]. The collected data's variable names and definitions can be found in Appendices A and B. The open-source intelligence data collection effort began in 2023 and continues at present.

Research Questions Approached:

RQ1. Is there a statistically significant difference in the distribution of cybersecurity impacts across the Confidentiality, Integrity, and Availability (CIA) pillars between non-profit organizations that align with the DHS CISA 16 Critical Infrastructure Sectors and those that operate outside these sectors?

RQ1a. Do non-profits operating outside the DHS CISA 16 Critical Infrastructure Sectors experience a disproportionately higher rate of availability impacts compared to sector-aligned non-profits?

RQ1b. Are confidentiality and integrity impacts more prevalent among non-profits aligned with DHS CISA Critical Infrastructure Sectors than among those outside the sector classification?

One area of focus and purpose of this study is to evaluate whether alignment with the sixteen DHS CISA Critical Infrastructure Sectors framework is associated with different cybersecurity impact profiles across the Confidentiality, Integrity, and Availability (CIA) pillars among non-profit organizations, using empirically observed incident data. Descriptive statistics are collected as part of the ongoing cyber incident research and awareness executive and include variables such as the date/year of the incident, the non-profit's name (if disclosed), country location, DHS CISA-aligned category, additional sector (if applicable), attack type, the CIA triad pillar affected, the source of OSINT information, the TAG cyber taxonomy, and whether other non-profits were targeted in the attack. All data were collected into a Microsoft Excel spreadsheet, with formatted columns and constraints on the fields with categories for selection. The procedures and strategy employed were outlined previously as part of a feasibility paper that demonstrated that this type of data collection and recording is possible [1]. In order to maintain reliability, as entries were identified for inclusion in the dataset, the information was initially evaluated by the researcher to determine if it was not a duplicate, then reviewed by a more senior analyst to ensure it was a valid entry. Proper taxonomy elements were selected and then finally evaluated by the

principal investigator. As new student researchers were added to contribute to the dataset, each was trained with a standardized manual that outlined the process and steps to be followed. The data were analyzed using IBM Statistical Package for the Social Sciences (SPSS) Statistics 28.0.1.1(15) software and Microsoft Co-Pilot Chat, feeding the Excel file for analysis, and the Co-Pilot data were further validated and double-checked. However, the investigation into finding incidents against the non-profit sector, as well as the process to catalog and categorize the incidents, did not include the use of any generative artificial intelligence (GenAI) tools. Finally, the data were summarized and combined with the previous running log of incidents, and it was stamped as a version twice a year.

4. Results

From the original version of the collected dataset, which was considered version 1.0 starting in 2023, the number of records has grown from $N = 168$ to the latest version 3.0, with $N = 463$, finalized in September 2025. For the analysis of the results in this research, all data assessed were combined into a total of 463 data records.

The following descriptive statistics are provided as a result of the current analysis in this longitudinal data collection effort.

This research effort involved manually assigning an attack type when cataloging the discovered cyber incident or attack; the types are outlined in Appendix B. Table 1 lists the top three attack types that were assigned, including breach, ransomware, and distributed denial-of-service attacks.

Table 1. Top attack types.

Attack Type	Count
Breach	303
Ransomware	127
Distributed Denial of Service (DDoS)	29
Malware	16
Denial of Service (DoS)	10
Defacement	7
Fraud/Scam	5
Advanced Persistent Threat (APT)	5

The details of each of the discovered and identified cyberattack incident were connected to the CIA triad, and it was possible to include multiple prongs of the triad. The count and percentage of the currently collected records are provided in Table 2. The majority of the attack incident details focus on the impact of the occurrence, specifically on the availability point.

Table 2. Confidentiality–Integrity–Availability (CIA) triad impacts.

CIA Impact	Count	Share (%)
Availability (A)	310	67.0
Confidentiality (C)	256	55.3
Integrity (I)	103	22.2

The prevalence (% of the CIA pillar within the attack type) and relative risk (RR compared to all other attack types) for the top-listed attack types are shown in Table 3.

Table 3. Attack types in relation to CIA impacts via prevalence.

Attack Type	CIA Pillar	Prevalence (%)	Relative Risk vs. Others
Ransomware	Availability	76.4	1.20
Breach	Confidentiality	61.0	1.32
DDoS	Availability	100.0	1.54
Defacement	Integrity	80.0	3.70
Fraud/Scam	Integrity	100.0	4.60
Breach	Availability	60.3	0.77

The CIA pillar of availability led the way in harming non-profit organizations, being associated with 67% of the attacks. Both ransomware and DDoS attacks, as categorized by the TAG taxonomy association, contributed to such incidents. Organizations should take mitigating steps to recover from disruptions, such as maintaining rigor in their backup/restore procedures, minimizing risk through network segmentation, and reducing DDoS potential.

The CIA pillar of confidentiality accounted for 55% of the breaches type of attacks. Organizations should emphasize proper data governance, strong access controls, and greater monitoring of end-user activity and anomalies. Regarding the TAG taxonomy, a focus is placed on data security, website browsing, and email security, which are types of activity for defending against these attack types.

While the CIA pillar of integrity was less common overall in the data, the highest categories referred to both defacement and fraud/scam. Consideration of using web application filtering, as well as content management system hardening and validating or verifying authentication practices, will contribute to anti-scam control.

As part of the classifying process, cyber incidents were reported as connecting to one or more CIA triad elements. The results of this analysis, combining the co-occurrence patterns, are outlined in Table 4. The availability pillar constitutes the largest share (41.5%) of attacks discovered against the non-profit sector. Of the CIA pillar categories, including combining them as part of the non-profit cyberattack type classification, availability (A) was most common in stand-alone systems, followed by confidentiality (C), and then with the combination of confidentiality (C) and availability (A). In many incidents, more than one component of the CIA triad was affected by the attack.

Table 4. CIA co-occurrence patterns (how CIA pillars combine).

CIA Combination	Count	Share (%)
A	192	41.5
C	90	19.4
C + A	78	16.8
C + I	53	11.4
C + I + A	35	7.6
I	10	2.2
I + A	5	1.1

The TAG Infosphere provides professional research and advisory services to the cybersecurity community and has developed a taxonomy of elements for describing both cybersecurity businesses and attack-type defenses [16]. Table 5 details the results of analyzing the TAG taxonomy elements and provides the qualities of the most common attack types. Data security, network security, and web security were the three top attack type vectors identified in the latest NPCIR dataset.

Table 5. Top 10 TAG cyber taxonomy elements.

TAG Taxonomy Element	Count
Data Security	246
Network Security	124
Web Security	103
Enterprise IT Infrastructure	92
Email Security	80
Cloud Security	45
Mobility Security	32
Application Security	25
Endpoint Security	18
Encryption and PKI	11

In cataloging the location of the country where the non-profit organization was affected by a cybersecurity attack or incident, Table 6 demonstrates that the United States, the United Kingdom, and Germany were the top three countries where non-profits were reported as being affected by the issue.

Table 6. Top 10 countries of events.

Country (Event)	Count
United States	222
United Kingdom	46
Germany	40
France	28
Canada	26
India	9
Belgium	9
China	8
Switzerland	7
Australia	7

Of the DHS CISA critical sectors, the HEALTHCARE AND PUBLIC HEALTH SECTOR topped the list. This finding parallels what was found in an earlier feasibility study on the topic of documenting cyber incidents and attacks against the non-profit sector [Table 7] [1].

Table 7. Top 10 DHS CISA critical sectors targeted.

DHS CISA Sector	Count
Healthcare and Public Health Sector	122
Government Facilities Sector	93
Emergency Services Sector	17
Information Technology Sector	14
Communications Sector	14
Financial Services Sector	13
Transportation Systems Sector	9
Commercial Facilities Sector	5
Food and Agriculture Sector	4
Water and Wastewater Systems Sector	3

When encountering a non-profit organization that did not align with any of the DHS CISA critical infrastructure sectors, an additional field was provided for manually cataloging the incident. The top ten areas are listed in Table 8. Education, local government,

and humanitarian aid-based organizations topped the list of recorded incidents from the longitudinally collected attacks.

Table 8. Top 10 areas outside the 16 DHS CISA sectors (if applicable).

Area Outside DHS CISA Categories	Count
Education	93
Local Government	73
Humanitarian Aid	18
Charity	15
Faith-based or religious	13
Political Organizations	8
Support Democracy	5
Arts	5
Entertainment Services	5
Scientific Services	5

Table 9, outlining the collected cataloged years of incidents discovered, indicates an inconsistency in distribution, with the year 2024 having an overwhelming larger number. More data collection is necessary in order to possibly determine the statistical significance of the results, and this may be possible as this effort to catalog cyberattacks against the non-profit sectors continues.

Table 9. Catalog of years (top) of incidents collected.

Year of Incident	Count
2024	309
2023	43
2025	43
2020	22
2022	18
2019	4
2017	3
2012	3
2018	2

The incidents categorized as part of sectors outside the DHS CISA critical sectors were significantly more impacted by Availability (80.2%) compared to 55.3% for the sectors outside the DHS CISA critical sectors. The confidentiality and integrity pillars of the CIA triad were more often affected the DHS CISA sector critical infrastructure areas. It should be noted that 54 records of the dataset analyzed were excluded from the prevalence by group analysis, as the incidents aligned with organizations that were part of a DHS CISA critical sector with involvement in sectors outside the DHS CISA sectors.

The CIA pillar impacted differed significantly between the groups of non-profit categories within DHS CISA sectors, which is detailed in Tables 10 and 11. Namely, the DHS CISA sectors experienced attacks that aligned or focused on the confidentiality and integrity pillars. Those outside the DHS CISA sectors were harmed more frequently by cyberattacks on the availability pillar.

Table 10. CIA pillar prevalence by group (DHS CISA sector vs. outside DHS CISA categories).

Group	Confidentiality (%)	Integrity (%)	Availability (%)
DHS CISA Sectors (N = 217)	67.3	33.2	55.3
Outside DHS CISA Sectors (N = 192)	41.7	9.4	80.2

Table 11. Chi-square tests of independence (CIA pillars vs. group (sector vs. outside)).

CIA Triad Pillar	N	Chi2	Df	p-Value	Cramer's V
Confidentiality	409	26.0069101	1	<0.001	0.25216379
Integrity	409	32.2612092	1	<0.001	0.280852741
Availability	409	27.4674785	1	<0.001	0.259147929

In evaluating the top attack types among the years with the most recorded incidents, Table 12 demonstrates that the attack category of ransomware dropped from 27.5% (2024) to 18.6% (2025), while DDoS attack types increased from 5.8% in 2024 to 9.3% in 2025. Breach attack types remained consistently above 60% in the years 2023, 2024, and 2025.

Table 12. Total cyber incident volume and most common attack type.

Year	Total	Breach	Ransomware	DDoS	Breach (%)	Ransomware (%)	DDoS (%)
2023	43	26	11	2	60.5	25.6	4.7
2024	309	197	85	18	63.8	27.5	5.8
2025	43	27	8	4	62.8	18.6	9.3

5. Discussion and Implications

Non-profit organizations are valuable contributors to communities, states, and countries and can be found in any part of the globe. When there is an unmet need or gap that is not being served by either business or public agency sectors, a non-profit may be a valuable contributor. As such, these organizations make valuable contributions to communities and, in our present day, require protection against cyber threats just like any other organization. Since the inception of the NPCIR, requests to access the data for analysis have been received and fulfilled; many of the collected data requests come from non-profit technology or security leaders around the globe. This is important to this work, indicating there is a place for such a repository. It is envisioned that this repository, as it grows, will be requested more frequently and may become a routine addition to an organization's management of information technology security. Recognizing the importance of the potential for cyber risks to evolve into real attacks or breaches is an important area of awareness for non-profit organization leaders [17]. Currently, more than 50 requests for the publicly and freely available dataset resulting from the NPCIR project have been received, and it is hoped that the dataset will continue to grow and become a valuable asset to individuals and teams tasked with defending their non-profit organizations against attacks.

6. Conclusions

The Non-Profit Cyber Incident Repository (NPCIR) [18] project, which began with envisioning and feasibility, has now been collecting data in its third year. This longitudinal effort has produced three stamped versions of publicly available data. The latest finalized dataset, namely, version 3.0, was analyzed and summarized in this research. Cybersecurity curriculum and practices continue to refer to the CIA triad as a form of cataloging and understanding security risks. Availability was the pillar most affected by the identified

attacks against the non-profit sector globally. In the availability aspect, rendering systems and solutions unavailable or inaccessible aligns with Routine Activity Theory (RAT) when considering that ransomware was a top attack type identified, which seeks to obtain monetary means to release or make data and solutions available to those attacked. Additionally, Socio-Technical Systems Theory (STS), which pertains to the human aspects of computing, connects with the availability pillar as a top affected element, given that breaches can start with a human, and a decision to address attacks such as ransomware attacks will also require a human decision. The pillar of confidentiality loss was identified as second most common, but still substantial, at 55%, which concentrated as breach attacks. The integrity pillar was identified as affected in non-profit organizations but at a lesser value of 22%.

6.1. Limitations

During this evaluation, several limitations were identified. One included the strong possibility that not every cybersecurity or cybercrime incident is publicly disclosed and thus cannot be found via open-source intelligence research and therefore cannot be cataloged. Unlike other critical sectors, such as finance, energy and power, and water and wastewater sectors, the non-profit sector is not required to report its cybersecurity incidents. There is the potential that if a non-profit organization were breached, but did not report it or go public with the information, the incident would not have been collected in this project. In addition, given that this is an ongoing research effort that collects yearly longitudinal data, cyber incidents could be reported at a later date, increasing the value of yearly cyberattack events. Given the rapid and recent movement into generative artificial intelligence, this data and open-source intelligence collection effort did not utilize AI. All the cataloging and categorizing of the identified open-source intelligence data were conducted manually, by humans. This is a limitation in efficiency, but it does offer more hands-on and intimate analysis of the data and information. However, this can also be a limitation; with AI, there is the potential for greater efficiencies and expedience. As shown in Table 9, there was an uneven number of cyber incidents across the years, with an increase in 2024 compared to other years. This limitation requires further evaluation going forward, with additional records continuing to be cataloged as of the time of the writing of this paper, which would provide more data to report back to the community. There is a limitation in understanding why there was uneven reporting, and additional longitudinal data collection is expected to continue going forward through the established process, which may provide further insights.

6.2. Future Research Directions

The findings of this study open several promising avenues for future research in the domain of non-profit cybersecurity and cyber threat intelligence. While the Non-Profit Cyber Incident Repository provides a valuable longitudinal dataset, further work is necessary to expand both the analytical depth and the methodological rigor of this research stream. Future studies should focus on enhancing the automation of data collection processes through the integration of artificial intelligence and machine learning techniques. The current manual OSINT-based approach, while thorough, may limit scalability and introduce potential human bias. Leveraging natural language processing and automated threat intelligence tools could improve both the volume and consistency of the collected data, while enabling the near-real-time monitoring of cyber incidents affecting non-profit organizations. There is a need to expand the geographical and sectoral diversity of the dataset. The current distribution of incidents is heavily concentrated in certain regions, particularly in North America and Europe. Future research should aim to incorporate under-represented regions such as Africa, South America, and parts of Asia to provide

a more globally representative understanding of cyber threats targeting non-profit organizations. Additionally, comparative studies across different types of non-profits (e.g., humanitarian, educational, faith-based) could yield more granular insights into sector-specific vulnerabilities. Finally, there is significant potential to examine the evolving role of emerging technologies, particularly generative artificial intelligence, in both offensive and defensive cybersecurity contexts. Future research should investigate how threat actors leverage AI-driven tools to target non-profits, as well as how non-profit organizations can adopt AI-enabled defenses to enhance resilience against cyber threats. In conclusion, expanding the NPCIR dataset, integrating advanced analytical methods, and deepening theoretical application will be essential steps toward building a more comprehensive and predictive understanding of cybersecurity risks in the global non-profit sector.

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Conflicts of Interest: The authors declare no conflict of interest.

Appendix A

Collected Dataset Variables.

Variable Name	Variable Definition
First Name Contributor	The first name of the person filling out the form.
Last Name Contributor	The last name of the person filling out the form.
Email	The email of the person filling out the form.
Category	Is the person a student, faculty /staff, visitor?
School/Organization of Contributor	What school or organization does the person belong to?
Unique Identifier of Incident	A unique title for the breach or hack must be given.
Date of Incident	The date of when the incident occurred.
Year of Incident	The year of when the incident occurred.

Variable Name	Variable Definition
Non-Profit/NGO Name	What is the NGO's name?
Non-Profit/NGO Countries	What country is the NGO located in?
Non-Profit/NGO Countries of Event	What country did the breach/hack take place in?
Threat Actor Type	Any individual or group that engages in malicious activities targeting digital systems, networks, or data often exploits vulnerabilities for various purposes; actors can range from cybercriminals seeking financial gain to nation-state actors pursuing geopolitical objectives, and their motivations and tactics vary widely based on their goals and resources.
DHS CISA Critical Sector Targeted	DHS CISA critical sectors refer to the 16 critical infrastructure sectors identified by the Cybersecurity and Infrastructure Security Agency (CISA), which is part of the U.S. Department of Homeland Security (DHS); these sectors represent areas of the economy that are considered vital to the United States, and their disruption or destruction would have a debilitating effect on national security, economic security, public health, or safety.
Areas Outside 16 DHS CISA Sectors	Additional sectors.
Attack Type	9 various attack types were identified; participants can only select one.
CIA Affected	CIA triad is a fundamental model in information security that focuses on three key principles: confidentiality, integrity, and availability. Confidentiality ensures data are kept private and accessible only to authorized parties, integrity maintains the accuracy and consistency of data throughout its lifecycle, and availability ensures that data and resources are accessible to authorized users when needed. This triad serves as a guiding framework for organizations to develop comprehensive information security policies, implement appropriate security controls, and evaluate their overall security posture.
TAG Taxonomy Element	TAG Cyber Taxonomy includes major categories that correspond to state-of-the-art cybersecurity approaches. The taxonomy serves as a foundation for TAG Cyber Research as a Service (RaaS), Content as a Service (CaaS), and Advisory.
Location of OSINT Knowledge Source	The source link that was used for the record.
Other Non-Profit/NGOs Targeted	Did this hack/breach affect any other organization?

Appendix B

Dataset Drop-Down Selection for Variables.

Variable Name	Variable Drop-Down Options
Threat Actor Type	Hacktivists Script Kiddies Cyber Criminals Organized Crime Advanced Persistent Threat (APT) Other

Variable Name	Variable Drop-Down Options
DHS CISA Critical Sector Targeted	Chemical Commercial Facilities Communication Critical Manufacturing Dams Defense Industrial Base Emergency Services Sector Energy Sector Financial Services Sector Food and Agriculture Sector Government Facilities Sector Healthcare and Public Health Sector Information Technology Sector Nuclear Reactors, Materials, and Waste Sector Transportation Systems Sector Water and Wastewater Systems Sector
Areas Outside 16 DHS CISA Sectors	Humanitarian Aid Education Food Security Support Democracy Political Organizations Scientific Services Technical Services Arts/Entertainment Services Faith-based or religious Veterans Organization Labor Organization Social Welfare Organization Foundations Charity
Attack Type	Breach Ransomware DoS DDoS Deface Virus/Malware SCAM Advanced Persistent Threats (APTs) State-Sponsored Other
CIA Affected	Confidentiality Integrity Availability

TAG Taxonomy Element	Application Security
	Attack Surface Management
	Authentication
	Cloud Security
	Data Security
	Email Security
	Encryption and PKI
	Endpoint Security
	Enterprise IT Infrastructure
	Governance, Risk, and Compliance (GRC)
	Identity and Access Management (IAM)
	Security Operations and Response
	Managed Security Services
	Mobility Security
	Network Security
	Operational Technology Security
	Security Professional Services
	Software Lifecycle Security
	Threat and Vulnerability Management
	Web Security
	Not Relevant
Location of OSINT Knowledge Source	The source link that was used for the record.
Other Non-Profit/NGOs Targeted	Did this hack/breach affect any other organization?

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